5 Thought

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Chapter overview

- Introduces the language of thought (LOT) hypothesis and evaluates the central arguments for it
- Explains and evaluates John Searle's Chinese room argument against the LOT hypothesis
- Introduces the Turing test and considers some of its strengths and weaknesses.

'Man is only a reed, the most feeble thing in nature, but he is a thinking reed'. So wrote the 17th-century philosopher and mathematician Blaise Pascal in his volume *Pensees* (§347). Thought is indeed one of our most notable attributes. But although we spend much of our time thinking, we rarely turn our cognitive faculties on themselves and consider their own nature. That is our task here.

Our discussion will revolve around two closely related questions. The first concerns the analysis of human thought. What does it take for creatures like us to have the kinds of thoughts that we do? This question is in many ways a question for psychologists and neuroscientists, but philosophers have had a great deal to say about it (as we shall see). A second – and more recognizably philosophical – question concerns the nature of thought as such. What are the essential features of thought? What would it take for an animal or an artificial neural network to think? DeepMind's AlphaZero can outplay humans in chess, and IBM's Watson has beaten the world's best players of the quiz show *Jeopardy!*, but are AlphaZero or Watson genuine thinkers?

We begin in Section 5.1 by distinguishing between thoughts of different kinds, and by asking what might distinguish thoughts from mental phenomena of other kinds. In Section 5.2 we explore the language of thought hypothesis (LOT). According to LOT, thoughts involve symbols in an internal representational system ('Mentalese'), and thinking involves nothing more than the manipulation of these symbols. LOT is influential, but it is also highly controversial. One challenge to LOT is that the manipulation of language-like symbols isn't necessary for thought; another challenge is that the manipulation of language-like symbols

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involve complex interactions between perception (looking at various items on the menu) and thought (wondering what a dish might contain or what it might taste like), but according to this view, no individual mental phenomenon can fall between perception proper and thought proper. Contemporary theorists, however, often view the border between perception and thought as nebulous and somewhat ill-defined, allowing that there might be phenomena that fall somewhere between paradigmatic perception on the one hand and paradigmatic thought on the other (Box: Alief: between thought and perception?).

Box: Alief: between thought and perception?

The Grand Canyon Skywalk is an enclosed viewing platform with a glass floor that extends 70 feet out into the Grand Canyon and allows one to look directly onto the canyon floor. Those who venture out onto it often experience vertigo. In some cases the vertigo is so overwhelming that people find themselves unable to walk to the middle of the viewing platform despite having paid for that privilege. How are we to understand this phenomenon? As Tamar Gendler (2008a, 2008b) has observed, appealing to what a person believes doesn't get us very far, for those who experience vertigo surely believe that the platform is perfectly safe. After all, why would they pay to step onto it if they had even the slightest doubt about its safety?

Gendler argues that in order to explain a person's reluctance to venture out onto Skywalk we need to posit a novel type of mental state that she calls 'alief'. Aliefs represent features of one's environment as having certain properties (e.g., as dangerous), they generate characteristic affective responses (e.g., feelings of anxiety and fear), and they activate various motor routines (e.g., muscle contractions associated with hesitation and retreat). Aliefs are akin to beliefs in that they guide the agent's behaviour, but unlike beliefs they are not subject to rational control. Reflecting on the evident safety of Skywalk might reinforce the belief that it is safe, but it won't suppress the queasy feeling in one's stomach or reduce the shaking of one's knees. Beliefs can be freely integrated with each other to form new beliefs (they are 'inferentially promiscuous'), whereas aliefs only enter into associative relations. Aliefs are akin to thoughts in certain ways, but in other respects they are more like perceptual states.

Gendler's arguments for the existence of aliefs has generated much discussion, and not everyone is convinced that the notion of alief must be added to our psychological arsenal. But whether or not the case for aliefs is ultimately successful, Gendler's work raises important questions about the nature of the border between thought and perception.

Further reading

Currie, G., and A. Ichino. 2012. Aliefs Don't Exist, though Some of Their Relatives Do. *Analysis Reviews*, 72(4): 788–98.
Mandelbaum, E. 2013. Against Alief. *Philosophical Studies*, 165(1): 197–211.
Nagel, J. 2012. Gendler on Alief. *Analysis Reviews*, 72(4): 774–88.

Some reasoning is automatic, intuitive, and fast. Reasoning of this kind has much in common with perception and occurs mostly unconsciously (although its results are typically available to consciousness). But reasoning can also be controlled, reflective, and slow. Psychologists often refer to the contrast between reasoning that is automatic, intuitive, and fast and reasoning that is controlled, reflected, and slow as the distinction between 'system 1' reasoning and 'system 2' reasoning. However, it is controversial whether these two modes of reasoning involve distinct systems or whether they involve using a single suite of systems in distinct ways.[1] Arguably the distinction between these two modes of reasoning is not hard and fast but instead marks out two ends of a continuum, with many instances of thought falling somewhere between the purely automatic and the purely controlled.

The contrast between automatic and controlled thought is most clearly revealed by cognitive illusions. These are cases in which our intuitions about a problem are at odds with our reflectively endorsed judgments. Consider the well-known *bat-and-ball problem* (Kahneman & Frederick 2002):

> *Bat and Ball Problem*: A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?

Most people's intuitive response is that the ball costs 10 cents. However, on reflection, it's evident that the ball must cost 5 cents and the bat $1.05. (If the ball cost 10 cents and the bat costs $1.00 more than the ball, then the bat would cost $1.10 and together they would cost $1.20.) What's striking, however, is that we remain primed to provide the wrong answer even when we know that it's wrong. In the same way that one's experience of a visual illusion is not penetrated by one's knowledge that it's an illusion, so too one's intuitive response to the bat-and-ball problem is not penetrated by one's reflective understanding. We share the capacity for automatic thought with the members of many other species, but the capacity for controlled thought may be a distinctively human trait. (We return to this theme in Section 5.4.)

5.2 The language of thought hypothesis

How is the capacity for thought to be explained? The most influential answer to this question is that thought involves the manipulation of sentences in a kind of inner language, what is sometimes called 'Mentalese'. Mentalese is not a public language in the way that English, Mandarin, or Portuguese is – instead, it's a purely private language. The idea that thought involves an inner language dates back to the Middle Ages but was given its canonical defence by Jerry Fodor (1935–2017) in his 1975 book *The Language of Thought*. The view (or at least close relatives of it) trades under a number of names (e.g., 'the representational theory of the mind', 'the computer model of the mind', 'the computational theory of the mind', 'the symbol-system hypothesis', 'sententialism'), but I will refer to it as the 'language of thought hypothesis', or LOT for short.

Central to LOT is the idea of a symbol (or representation). Symbols have both syntactic and semantic properties. A syntactic (or formal) property is a property that a purely physical system could be sensitive to and which is correlated with a semantic property. Some of

differences between the names 'Mrs. Kahn' and 'Mr. Jones'. As John Haugeland (1985: 106) memorably put it, 'If you take care of the syntax, the semantics will take care of itself'.

That, in essence, is how LOT accounts for thought. The basic idea is that thoughts are symbol structures, and thinking involves the manipulation of these structures on the basis of their syntactic properties. Does thinking *feel* like it involves symbol manipulation? Probably not, although it's not exactly clear what symbol manipulation would feel like. But in any case, the issue is irrelevant, for LOT doesn't make any claims about what it feels like to think. Instead, LOT is an account of the machinery of thought.

Advocates of the LOT hypothesis have different conceptions of the scope of the account. One approach – we might call it the 'essentialist view' – treats LOT as an account of the nature of thought as such (see, e.g., Davies 1992, 1998). According to the essentialist, no creature could be a thinker unless it had some version of a language of thought. By contrast, the non-essentialist version of LOT (as defended by Fodor) treats it as an empirical claim about the nature of human thought and allows that thought could occur in the absence of Mentalese. The contrast between these two views has important consequences. Most obviously, it bears on the analysis of non-human thought. If the essentialist version of LOT were right, then the question of whether a certain kind of AI system qualifies as a thinker would need to be addressed by asking whether it has a language of thought. By contrast, if the LOT hypothesis were regarded as only one possible way in which thought can be realized, then we couldn't draw any conclusions about whether an AI system was a thinker from its failure to employ language-like representations. We will focus on the non-essentialist version of LOT, for it is this form of the view that has had the most influence.

Having seen what the LOT hypothesis says, let us consider what might be said in favour of it. The most influential arguments for LOT are Fodor's arguments from productivity and systematicity. The productivity and systematicity of thought are both aspects of what we earlier referred to as its open-ended nature. Thought is productive in that there don't seem to be any principled limits on the range of thoughts that we can entertain. Fodor puts the point as follows:

> The thoughts that one actually entertains in the course of a mental life comprise a relatively unsystematic subset drawn from a vastly larger variety of thoughts that one could have entertained had an occasion for them arisen. For example, it has probably never occurred to you before that no grass grows on kangaroos. But, once your attention is drawn to the point, it's an idea that you are quite capable of entertaining, one which, in fact, you are probably inclined to endorse.
>
> (Fodor 1985: 89)

The systematicity of thought has two faces. The first is that the capacity to think certain thoughts comes along with the capacity to think other thoughts. As Fodor points out, we don't find people who can think MARY LOVES JOHN but cannot think JOHN LOVES MARY. The second face of systematicity is that inferential capacities are systematically related to each other. For example, we don't find people who can infer *P* from *P&Q* but cannot infer *Q* from *P&Q*.[3]

Fodor argues that the productivity and systematicity of thought are best explained by supposing that thought is compositional. The model here is provided by our linguistic capacities. Linguistic capacities are systematic, for if you know what 'Mary loves John' means, then you

systematicity and productivity. But does LOT provide the only (or even the best) explanation of this fact?

One challenge to LOT derives from connectionist models of cognitive architecture. These models were first developed in the 1980s and over time have given rise to the deep-learning networks that drive voice- and face-recognition systems (Buckner 2019; Lake et al. 2017; LeCun et al. 2015). Instead of a central processor and symbolic representations that are stored in memory, a connectionist network involves layers of nodes that bear weighted connections to one another. Learning changes the weights on these nodes, and computation in a connectionist network involves activating the network nodes, with a node's activation level being dependent upon the weighted activations of the nodes to which it is connected. Crucially, in a network processing is both distributed and parallel – hence connectionist architectures are known as parallel distributed processing architectures (Rumelhart et al. 1987). By contrast, in a LOT-based architecture computation is typically localized and serial. With the rise of connectionism, the LOT approach to cognition has become known as the classical approach to cognition.

Connectionism suggests an alternative account of thought to that provided by LOT. Because representation in a network is distributed across the network, a network can represent THE CAT SAT ON THE MAT and THE CAT CHASED THE DOG without these representations having any common element. The overall state of the network might encode the content THE CAT SAT ON THE MAT, but there may be no component of the network that represents the cat (or the mat, for that matter). Although the *contents* of thought will have constituents – the thought THE CAT SAT ON THE MAT 'contains' the concepts CAT, SAT, and MAT – the structure of thoughts themselves may not reflect the structure of their contents. In a LOT-based architecture, by contrast, systems that represent THE CAT SAT ON THE MAT and THE CAT CHASED THE DOG will have a common element; namely, the symbol CAT.

The mere fact that connectionism is an alternative to the classical LOT-based approach doesn't itself undermine the LOT hypothesis, for it might be argued that the LOT hypothesis does a better job of explaining the crucial features of thought than connectionist accounts do. Indeed, that is precisely the claim made by Fodor and a number of his colleagues (Fodor & Pylyshyn 1988; Fodor & McLaughlin 1990). In order to account for the systematicity of thought, they argued, there must be a higher level of description of any connectionist network according to which it can be thought of as implementing a language of thought. In a nutshell, Fodor and his colleagues responded to the connectionist challenge with a dilemma: either connectionist networks are mere implementations of a LOT-based architecture (in which case they aren't genuine rivals to LOT), or they are genuine rivals to LOT but fail to account for the systematicity of thought. Let's call this the *systematicity argument*.[5]

Evaluating the systematicity argument requires turning the spotlight on systematicity itself. Let us grant that mature human thought is systematic. In other words, let's grant Fodor's claim that you just don't find people who can think MARY LOVES JOHN but not JOHN LOVES MARY. Let's also grant that connectionism as such lacks the resources to explain this fact. (Although connectionist networks *can* exhibit systematicity, there is nothing in their

Box: The scaffolding of thought

If language makes a cognitive contribution to thought, then the question arises as to *how* it makes this contribution.

One possibility is that the use of symbols enables a creature to transform problems that are too difficult for it to solve into soluble ones. An intriguing example of the transformative power of symbols is provided by a study of chimpanzees that were trained to use symbols (plastic tags) to represent the relations of sameness and difference (Thompson et al. 1997). For example, a pair of cups might be associated with a red triangle to indicate that they are objects of the same kind, whereas a cup and a shoe might be associated with a blue circle to indicate the difference between them. After training, chimpanzees that had been trained in the use of these symbols – and only those chimpanzees – were able to use the tags to appreciate relations of higher-order sameness and difference. In other words, they were able to appreciate that two pairs (such as cup-cup and cup-shoe) instantiate the relation of difference, since the first pair exhibits the sameness relations and the latter pair exhibits the difference relation. The authors suggest that the symbols enabled the chimpanzees to perform this task because by visualizing the tags, they could transform a higher-order task (identifying how relations are related) into a first-order task (determining how symbols are related). As Andy Clark puts it, 'Experience with external tags and labels thus enables the brain itself . . . to solve problems whose level of complexity and abstraction would otherwise leave us baffled' (2006: 294).

Of course, natural language isn't the only tool that we have at our disposal for scaffolding thought. We restructure our cognitive niche so as to make thinking easier, more productive, and less fragile. Many of us do our best thinking with the aid of artefacts: a pen and paper, an abacus, or a laptop (see Chapter 7). We also learn how to think by being guided by others (Sterelny 2012). In his *Critique of Pure Reason* (A134/B173–74), Kant referred to the examples that the young child is given as 'the *gängelwagen* of thought', where a *gängelwagen* is a walking frame or go-kart that is harnessed to an infant in order to help them learn to walk. The social scaffolding of thought is not restricted to infancy and childhood but continues throughout life. The notion of thought tends to evoke images of the solitary scholar confined to their study, but in fact the vast majority of human thought surely occurs in social settings. We often do our best thinking in conversation with others, relying on another person's prodding and probing to keep our own thoughts on track.

Further reading

Clark, A. 2006. Material Symbols. *Philosophical Psychology*, 19: 291–307.
Dove, G. O. 2014. Thinking in Words: Language as an Embodied Medium of Thought. *Topics in Cognitive Science*, 6: 371–89.
Heyes, C. 2018. *Cognitive Gadgets: The Cultural Evolution of Thinking*. Harvard University Press.

think that? Searle himself is committed to the view that a thinking thing can be composed of unthinking components, for he takes thought to emerge from the biological operations of the brain, and the components from which brains are constructed are not themselves thinkers. Indeed, unless thought is a primitive feature of the physical world, it *must* be the case that thinking systems are built out of things that aren't thinkers!

Second, Searle claims that the Chinese room argument can be easily modified to accommodate the systems reply:

> Let the individual internalize all of these elements of the system. He memorizes the rules in the ledger and the data banks of Chinese symbols, and he does all the calculations in his head. The individual then incorporates the entire system. There isn't anything at all to the system that he does not encompass. . . . All the same, he understands nothing of the Chinese, and a fortiori neither does the system, because there isn't anything in the system that isn't in him. If he doesn't understand, then there is no way the system could understand because the system is just a part of him.
>
> (Searle 1980: 419)

This response is no more convincing than the first, for it commits a compositional fallacy. The fact that an individual has some property (failure to understand) doesn't imply that any component of that individual must also have that property. (I am the author of this book, but I don't share that property with any of my components.)

Let's turn now to the first premise of the Chinese room argument and ask whether the scenario outlined by Searle involves symbol manipulation. It certainly seems to. After all, Searle stipulates that the person in the room is given messages written in Chinese characters, and the instructions that they are required to follow direct them to produce messages written in Chinese characters. But should we grant that these symbols are genuine Chinese characters? After all, not everything that looks like a Chinese character is a genuine Chinese character. (The movements of an ant on a beach might produce marks that resemble the Chinese characters for telephone, but those marks won't constitute genuine Chinese characters.) In order to be a Chinese symbol – indeed, in order to be a symbol of any kind – the relevant marks must be hooked up to the environment in certain ways. But for all Searle says, the symbols in the Chinese room would lack any connection to the environment.

There are various ways in which we might connect the Chinese room's symbols up to its environment. We could equip it with perceptual systems, so that its symbols are appropriately activated by perceptual input. (For example, we could ensure that perceiving a telephone activates the symbols for telephone.) We could also equip the Chinese room with motor systems, so that its representations are able to drive its behaviour. (For example, we could set things up so that forming a representation with the content 'I need a telephone' leads to telephone-seeking behaviour.) In effect, we could embed the Chinese room in a robot, an entity that both perceives and acts on its environment. Having done that, the symbols in the Chinese room would be related to features of the environment in the ways in which they need to be in order to mean anything. But – and here is the crucial move – once the Chinese room has been embedded in a robot and located in some portion of the world, the intuition that it is devoid of thought starts to loosen its grip.

One of the first people to wrestle with this question was the British mathematician, philosopher, and codebreaker Alan Turing (1912–54). Turing (1950) suggested that the question of whether a machine can think should be replaced with another question: 'Can a machine win the imitation game?' The imitation game involves three individuals: an interrogator, a human being, and a machine whose status as a thinker is under consideration. The identities of the human being and machine are kept hidden from the interrogator, who is allowed to put questions of any kind to them. The machine wins the game if, after an extended session of questioning, the interrogator is unable to reliably determine which of the players is the human and which is the machine. This test for machine thought is commonly referred to as 'the Turing test'.

Turing leaves a number of crucial parameters of his test unspecified. For instance, he doesn't say how many questions the interrogator is allowed to ask before the game concludes, nor does he say what kind of background knowledge the interrogator is allowed. However, the central idea behind the test is clear enough: a machine can be said to think (or be intelligent) if its conversational skills are equal to ours.

This is in many ways an attractive proposal, for there are no limits on the types of questions that the interrogator can ask. The interrogator can ask why a certain joke is funny, what a particular proverb means, or what the history of the 20th century might have looked like if the combustion engine had not been invented. Although Turing himself suggested that by 2000 computers would be able to play the imitation game so well that 'an average interrogator will not have more than a 70% chance of making the right identification after five minutes of questioning', to date no AI system has passed the Turing test, and few qualified commentators expect it to be passed any time soon (Box: Winning the imitation game?).

Box: Winning the imitation game?

In 2014 a chatbot called 'Eugene Goostman' fooled 10 of the 30 judges in a Turing test competition, who took themselves to be conversing with a 13-year old Ukrainian boy. Although Eugene Goostman was widely described as having 'passed the Turing test', it is doubtful that Turing would have seen it that way, for Eugene Goostman's capacity for intelligent dialogue was severely limited. Consider the following conversation between Eugene Goostman and the computer scientist Scott Aaronson (2014):

Scott: Which is bigger, a shoebox or Mount Everest?
Eugene: I can't make a choice right now. I should think it out later. And I forgot to ask you where you are from. . . .
Scott: How many legs does a camel have?
Eugene: Something between 2 and 4. Maybe, three? :-))) By the way, I still don't know your specialty – or, possibly, I've missed it?
Scott: How many legs does a millipede have?

agency of another individual (such as a programmer). If a machine 'can only do what we tell it to do' – in Turing's paraphrase of Lovelace – then it cannot truly be said to think.

The issues raised by Lovelace's objection are complex, and we cannot do justice to them here. What we can say, however, is that any machine that is able to pass the Turing test is almost certain to surprise its designers, and in that sense won't do 'only what it is told to do'. An illustration of the originality of AI programs is provided by AlphaZero, which has beaten the world's best players in the Chinese game of Go. AlphaZero was successful not because of the instructions that it had been given but because it discovered novel strategies by playing against itself and exploring a vast space of possible strategies. Indeed, what most impressed Go experts wasn't AlphaZero's level of performance but its playing style. As world champion Ke Jie remarked: 'After humanity spent thousands of years improving our tactics, computers tell us that humans are completely wrong. . . . I would go as far as to say not a single human has touched the edge of the truth of Go' (quoted in Dou & Geng 2019).

A third objection to the Turing test involves consciousness. If thought requires consciousness, and if a machine could pass the Turing test without being conscious, then, the objection runs, the Turing test is not a good test for thought.

Could an unconscious system pass the Turing test? That may be possible in principle, but as we saw in connection with Blockhead, we need to distinguish what is possible in principle from what is possible in practice. Perhaps unconscious machines simply aren't up to the job, and only a system that was endowed with consciousness could generate intelligent responses in real time. At this point in time, we simply don't know. Although consciousness seems to be required for intelligent behaviour in human beings, many of the leading accounts of thought (such as LOT) make no reference at all to consciousness, and it is not uncommon for theorists to assume that consciousness and intelligence have very little to do with each other.[11] The issue is further complicated by the fact that there is little agreement about what criteria we should use for ascribing consciousness to machines (see Chapter 11). In short, no one really knows whether an unconscious machine could actually pass the Turing test.

Stepping back from the Turing test for a moment, what should we say about the relationship between thought and consciousness? Is unconscious thought possible, or should we reserve the term 'thought' for conscious phenomena? The paradigmatic instances of thought with which we began this chapter – deciding what to have for dinner, wishing that the rain would stop, realizing that an argument is invalid – are all conscious states. Even Turing recognized that there is an experiential aspect to thought, confessing that if he had to define thought, he would probably describe it as a 'sort of buzzing that went on inside my head' (quoted in Copeland 1999). But it is one thing to grant that paradigmatic thoughts are conscious and quite another to assume that thoughts *must* be conscious. Certainly a great deal of cognitive processing takes place outside of consciousness. Suppose that one has been puzzling over a problem for days; suddenly, the solution pops into one's mind. The solution itself was conscious, but the cognitive processes – thoughts, as we might call them – that gave rise to it were not. And even if human thought is essentially tied to consciousness, there may be forms of thought that are independent of consciousness.

Arguably, the most important challenge to the Turing test concerns the fact that it is a one-way test: failing it doesn't tell us anything about a machine's status as a thinker. This

Further reading

Louise Antony's entry on 'Thinking', in B. McLaughlin, A. Beckermann, and S. Walter (eds.), in *The Oxford Handbook of Philosophy of Mind* (OUP, 2009), examines a number of aspects of thought that are not covered in this chapter, as does my own *Thought: A Very Short Introduction* (OUP, 2013). Good entry points into the debate about what distinguishes thought from other faculties can be found in Elisabeth Camp's 'Putting Thoughts to Work: Concepts, Systematicity, and Stimulus-Independence', *Philosophy and Phenomenological Research*, 78(2): 275–311 (2009); Daniel Weiskopf's 'The Architecture of Higher Cognition', in M. Sprevak and J. Kallestrup (eds.), *New Waves in Philosophy of Mind* (Palgrave, pp. 242–61, 2014); and Jakob Beck's 'Marking the Perception-Cognition Boundary: The Criterion of Stimulus-Dependence', *Australasian Journal of Philosophy*, 96(2): 319–34 (2018).

Tim Crane's *The Mechanical Mind* (2nd ed., Routledge, 2003) and J. Christopher Maloney's *The Mundane Matter of the Mental Language* (CUP, 1989) provide excellent introductions to the language of thought. Fodor himself wrote two book-length defences of the language of thought hypothesis – *The Language of Thought* (1978) and *LOT 2: The Language of Thought Revisited* (2008) – but the most accessible introduction to his position can be found in the appendix to his book *Psychosemantics* (1987), titled 'Why There Still Has to Be a Language of Thought'. For two of the many variants on the standard version of LOT, see Susan Schneider's *The Language of Thought: A New Philosophical Direction* (MIT, 2011), and N. Goodman, J. Tenenbaum, and T. Gerstenberg, 'Concepts in a Probabilistic Language of Thought', in E. Margolis and S. Laurence (eds.), *The Conceptual Mind: New Directions in the Study of Concepts* (MIT Press, pp. 623–54, 2015).

Dennett's reservations about the LOT hypothesis can be found in his papers 'A Cure for the Common Code', in *Brainstorms* (MIT Press, 1981) and 'Mother Nature versus the Walking Encyclopedia: A Western Drama' in W. Ramsey, S. P. Stich, and D. E. Rumelhart (eds.), *Developments in Connectionist Theory. Philosophy and Connectionist Theory* (Lawrence Erlbaum, pp. 21–30, 1991). Good introductions to the contrast between connectionism and the classical LOT-based accounts of cognition can be found in William Bechtel and Adele's Abrahamsen *Connectionism and the Mind: Parallel Processing, Dynamics and Evolution in Networks* (2nd ed., Blackwell, 2002), especially pp. 341–43, and Andy Clark's *Mindware* (OUP, 2014), especially pp. 87–89. One of the most influential connectionist responses to the systematicity argument is to be found in Paul Smolensky's 'On the Proper Treatment of Connectionism', *Behavioral and Brain Sciences*, 11: 1–74 (1988).

Stimulating accounts of what is distinctive about human thought can be found in Michael Tomasello's *The Cultural Origins of Human Cognition* (Harvard University Press, 1999); Kim Sterlney's *Thought in a Hostile World* (Blackwell, 2003); and Cecilia Heyes's *Cognitive Gadgets: The Cultural Evolution of Thinking* (Harvard University Press, 2018).

John Searle's Chinese room argument can be found in his paper 'Minds, Brains, and Programs', first published in the journal *Behavioral and Brain Sciences*, 3, 417–57 (1980) and reprinted many times since. A collection of papers responding to the argument can be found in John Preston and Mark Bishop (eds.), *Views into the Chinese Room: New Essays on Searle and Artificial Intelligence* (OUP, 2002).

concerns the claim that connectionist models are more neurally plausible than their LOT-based rivals. For an excellent discussion of this issue see Stinson (2018).

6 An analogous point applies to human infants, who can presumably think I WANT MILK but probably can't think MILK WANTS ME.

7 For more on this theme, see Bermúdez (2003), Carruthers (2002, 2004), Rescorla (2009), and Penn et al. (2008).

8 See Copeland (1993, chap. 6) for an insightful defence of the systems reply.

9 There are some delicate issues here, for it might look as though the manipulation of symbols on the basis of their formal (i.e., syntactic) properties makes causal relations with the outside world irrelevant. (Consider again Haugeland's claim: 'If you take care of the syntax, the semantics will take care of itself'.) But even if the syntactic properties of a system ensure that the semantics takes care of itself, nothing counts as a symbol unless it has semantic properties.

10 In a radio interview that Turing gave on the question of machine thought, he remarked: 'I am not saying at present either that machines really could pass the test, or that they couldn't. My suggestion is just that this [i.e., "Can machines pass the Imitation game?"] is the question we should discuss. It's not the same as "Do machines think?", but it seems near enough for our present purpose, and raises much the same difficulties' (quoted in Copeland 1999).

11 Fodor (1998: 73) once remarked: 'I try never to think about consciousness. Or even to write about it'.